Stream Temperature and Salmon Returns in Issaquah Creek: A 29-Year Observational Study

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Abstract—Adult Chinook and Coho returns to Issaquah Creek, Washington, vary widely among years. This study tested whether warmer creek temperatures during biologically defined life-stage windows were associated with fewer returning adults and whether a proposed snowpack-to-flow-to-temperature pathway was detectable. Because continuous long-term water-temperature records are unavailable, daily water temperature for 1995–2025 was reconstructed with ridge regression using regional air temperature and seasonal terms, calibrated to 431 King County grab samples. Leave-one-year-out RMSE was 0.788 °C overall and 0.70–0.97 °C within the three biological windows. Five hypotheses were specified before confirmatory testing and evaluated with Spearman correlations, permutation tests, bootstrap confidence intervals, and Holm correction for multiple comparisons. None of the three salmon-temperature hypotheses remained supported after correction. Chinook returns showed the strongest pattern: warmer migration-window temperature was associated with fewer adults ($\rho = -0.400$; raw $p = 0.032$; Holm-adjusted $p = 0.095$), and the direction remained negative across all prespecified robustness checks. Coho relationships were weaker. April 1 snowpack was not associated with late-summer flow, while higher summer flow was associated with cooler modeled water ($\rho = -0.417$; Holm-adjusted $p = 0.039$). These results support a local flow-temperature relationship but not the full proposed snowpack-to-flow-to-temperature-to-salmon pathway.

I. INTRODUCTION

Pacific salmon experience temperature throughout freshwater migration, spawning, and juvenile development. Warmer water increases metabolic demand and can reduce habitat quality or increase disease risk [1]–[3]. In the Lake Washington basin, warm conditions have also been linked to prespawning mortality and migration difficulty in sockeye salmon [4], [5]. However, annual adult return abundance is more complicated than short-term physiological stress because returning fish have already passed through freshwater, estuarine, and marine life stages.

The study began with a simple mechanistic hypothesis: lower spring snowpack might reduce late-summer flow, lower flow might allow the creek to warm, and warmer water might be associated with fewer salmon returning. Rather than assume that chain was true, each link was tested separately. Recent work at the same hatchery found that Chinook and Coho spawning timing has shifted despite warming, showing that hatchery selection and other biological processes can complicate simple thermal expectations [6]. The present study therefore focused on a narrower question: are temperatures during specific life stages associated with annual adult returns, and are the proposed snowpack-flow-temperature links visible in the same record?

Life-stage windows were used instead of annual mean temperature because an annual average would include months when the fish or life stage of interest is not exposed to the creek. The adult windows cover the main hatchery-return periods for Chinook and Coho, while the Coho rearing window tests freshwater conditions two years before adult return. This design does not claim that every fish experienced the same temperature. It asks whether years with warmer conditions during a biologically relevant period also tended to have different adult return counts.

II. METHODS

A. Data and biological windows

Annual adult Chinook (Oncorhynchus tshawytscha) and Coho (O. kisutch) returns to the Issaquah Hatchery were obtained from Washington Department of Fish and Wildlife records [7]. The primary salmon analyses used 29 return years, 1997–2025, and excluded jacks. Environmental data included King County water-temperature grab samples at Issaquah Creek station 0631 [8], NOAA daily air temperature [9], USGS daily mean discharge near the mouth of Issaquah Creek [10], and April 1 snow-water equivalent (SWE) from NRCS Stampede Pass SNOTEL [11].

The return series is a hatchery-trap count rather than an estimate of total watershed abundance. Chinook adult counts in the analyzed record ranged from roughly 1,800 to 13,400, while Coho ranged from roughly 360 to more than 35,000, emphasizing how much year-to-year variability a freshwater predictor must explain. Adult count without jacks was retained as the primary outcome, while adults-plus-jacks was reserved as a robustness check rather than changing the outcome after seeing the results.

Before confirmatory testing, three salmon-temperature hypotheses were fixed around life-stage windows: Chinook adult migration (Aug. 15–Sept. 30 of the return year), Coho adult migration (Sept. 15–Oct. 31), and Coho juvenile/rearing exposure (June 1–Sept. 30 two years before return). Two physical hypotheses tested April 1 SWE versus July–September mean flow and June–September flow versus modeled water temperature. Longer migration windows were reserved as sensitivity checks and could not replace the primary windows after results were known.

B. Daily water-temperature reconstruction

Issaquah Creek has no continuous measured daily temperature record covering the return series. A ridge-regression proxy was therefore calibrated to 431 grab samples on 415 dates from 1995–2025. The primary model (T2) uses daily air-temperature midpoint and range, trailing 3-, 7-, and 30-day air-temperature means, and annual and semiannual sine/cosine terms. Streamflow is deliberately excluded so that flow can later be tested independently against modeled temperature. A secondary model (T1) includes flow and was used only for sensitivity analysis.

T2 was evaluated by leaving out one year at a time. Overall held-out RMSE was 0.788 °C, MAE was 0.618 °C, and R^2 was 0.955. Within the three hypothesis windows, RMSE was 0.771 °C for Chinook migration, 0.696 °C for Coho migration, and 0.970 °C for Coho rearing. Each window outperformed a held-out monthly-climatology baseline. The ridge penalty was selected within the same

leave-year-out framework, so these error estimates are somewhat optimistic compared with fully nested validation.

The daily proxy was then averaged within each frozen biological window to create one temperature exposure value per analysis year. Predictor-range extrapolation was also audited for window days outside the calibration-data ranges. Flagged years were kept in the primary analysis, as specified in advance, and then removed in a sensitivity analysis. This avoided selectively discarding unusual years while still showing whether extrapolation drove any result.

C. Statistical analysis

The confirmatory statistic was Spearman rank correlation (ρ). Two-sided p-values used 100,000 random permutations, and 95% confidence intervals used 10,000 bootstrap resamples. Multiple testing was controlled with separate prespecified Holm families: three salmon-temperature tests and two physical-mechanism tests. A hypothesis was labeled supported only when the direction matched the prediction, Holm-adjusted $p < 0.05$, and required robustness checks did not reverse the direction.

For salmon hypotheses, descriptive regressions of log1p(adult count) on standardized temperature with HC3 robust errors were also fit; these described effect size but did not replace the rank-based decision rule. Prespecified checks included using T1, longer migration windows, adults plus jacks, excluding temperature-extrapolation years, removing the most influential regression year, and leaving out each year in turn. Because both temperature and returns can trend over time, A1 was also repeated after linear detrending of ranks, and a circular-shift diagnostic was used when residual autocorrelation crossed the protocol threshold. The full protocol, code, processed data, and saved outputs are available in the project repository.

To reduce researcher flexibility, the confirmatory protocol, windows, lags, multiplicity families, and required sensitivity checks were versioned before the Phase 7 salmon analysis. The implementation saves machine-readable result tables and validation metadata. A later repository audit reran the current pipeline and reproduced the substantive Phase 7 statistical tables exactly. The code, saved outputs, and validation scripts are retained so that the reported results can be checked against the implemented analysis.

D. AI-assisted tools

The original 24-page manuscript was written by the author. OpenAI ChatGPT was later used for code troubleshooting/review and editorial support in later revisions, including condensation of the author's original manuscript into this CJSJ format. For this three-page CJSJ version, the principal prompt was: "please make CJSJ-specific 3 page submission manuscript based on their official template, repos reports, V4 and V5 docs." The tool was ChatGPT (GPT-5.6 Sol). AI did not select the research question, hypotheses, biological windows, multiplicity rules, or interpretation criteria, and all numerical results were generated from and checked against repository outputs. A complete record of additional AI-assisted coding and revision prompts, with tool/version information, will be disclosed to CJSJ with the submission.

III. RESULTS AND DISCUSSION

None of the three salmon-temperature hypotheses met the corrected support threshold (Fig. 2). Chinook had the largest relationship: warmer Aug. 15–Sept. 30 temperature was associated with fewer adult returns ($\rho = -0.400$; 95% bootstrap CI -0.677 to -0.045). The raw permutation p-value was 0.032, but Holm adjustment increased it to 0.095. The Chinook result is therefore treated as statistically inconclusive rather than confirmed. The descriptive regression estimated about 21.5% fewer fitted adults per one-standard-deviation warmer migration window, but that estimate does not override the corrected hypothesis test.

The two Coho associations were weaker. Adult migration temperature was nearly unrelated to returns ($\rho = -0.044$; Holm $p = 0.821$), while the juvenile/rearing estimate was negative but uncertain ($\rho = -0.202$; Holm $p = 0.580$). The contrast is useful because the same basic design did not produce similar patterns for both species. Coho migrate later in cooler conditions, and hatchery Coho are yearlings, so the biological meaning of a creek rearing-temperature window may differ from that of a wild juvenile. These are possible explanations, not conclusions tested directly here.

A secondary Chinook model entered migration-window temperature and matched-window streamflow together. The standardized temperature coefficient remained negative ($\beta = -0.240$, $p = 0.049$), while the flow coefficient was near zero ($\beta = -0.035$, $p = 0.794$; adjusted $R^2 = 0.126$). For Coho, neither predictor was distinguishable from zero and adjusted R^2 was -0.041. These models are descriptive because they were not the confirmatory decision rule and cannot establish that temperature itself caused lower returns.

The physical pathway split in two (Fig. 3). April 1 SWE at Stampede Pass was not associated with July–September Issaquah Creek flow ($\rho = +0.137$; Holm $p = 0.477$). In contrast, higher June–September flow was associated with cooler modeled water ($\rho = -0.417$; 95% bootstrap CI -0.677 to -0.078; Holm $p = 0.039$). The flow-temperature direction is consistent with established stream-thermal relationships [12], [13]. The null snowpack result also changes interpretation of the proposed pathway: Stampede Pass is a regional mountain snow proxy, while Issaquah Creek is a lower-elevation Puget Sound watershed influenced strongly by rainfall and groundwater. A snow-dominant Cascade-basin intuition therefore does not necessarily transfer to this watershed [14].

Although A1 did not survive multiplicity correction, its direction was stable. Replacing T2 with T1 left ρ essentially unchanged (-0.400); the longer migration window gave -0.327, including jacks gave -0.391, excluding extrapolation-flagged years gave -0.443, and removing the most influential regression year gave -0.503. Across all 29 leave-one-year-out reruns, ρ remained negative (-0.503 to -0.347). After removing a linear calendar-year trend from both rank series, ρ was -0.352, suggesting that a shared linear trend does not fully account for the pattern, although nonlinear temporal structure and unmeasured confounding remain possible. The prespecified circular-shift diagnostic gave $p = 1/29 = 0.034$, but only 29 shifts are possible, so that value is coarse and was treated only as a sensitivity result.

The Chinook pattern is therefore interesting for its consistency rather than for crossing a significance threshold. The estimate survives reasonable changes in model form and sample composition, but the study is still observational

and small. This distinction matters because it would be easy to describe raw $p = 0.032$ or the negative regression interval as proof of a temperature effect. Under the analysis rule fixed before testing, A1 is not supported after family-wise correction. The appropriate conclusion is that the data identify a stable signal worth testing with a longer and more complete record.

Several limitations prevent causal interpretation. Water temperature is reconstructed rather than continuously measured; the adult-return record contains only 29 years; annual hatchery release counts, harvest, age composition, detailed ocean survival, and predation were not modeled; and the snowpack measure is not a basin-specific storage estimate. Statistical power is also limited: under the simulation used in the analysis, roughly $|\rho| = 0.57$ was needed for 80% power at the strictest salmon-family Holm threshold, while $|\rho| = 0.40$ had only about 39% power. Therefore, the nonsignificant salmon tests do not tightly rule out moderate ecological relationships. A7 also depends on how the five hypotheses are grouped for multiple-comparison correction: it passes the prespecified two-test mechanism-family correction but would not pass if all five hypotheses were treated as one family.

The larger ecological result is that the proposed four-step chain was too simple. The flow-temperature link is a local watershed-physics relationship. Adult returns, in contrast, integrate years of freshwater and marine exposure, hatchery practices, harvest, predators, disease, and cohort structure. The nearby Issaquah Hatchery phenology study by Quinn et al. [6] similarly shows that long-term warming does not translate into a single simple salmon response. The present study addresses abundance rather than spawn timing, but both point toward separating measurable thermal stress from the many processes that determine which adults eventually return.

IV. CONCLUSIONS

Across 29 years of Issaquah Hatchery returns, warmer migration-window temperature was moderately associated with fewer Chinook adults, but the association did not survive the prespecified family correction. Coho relationships were weak or uncertain. Higher summer flow was associated with cooler modeled water, while the regional snowpack proxy did not predict late-summer Issaquah Creek flow. The original snowpack-to-flow-to-temperature-to-salmon chain was therefore not observed intact. The main lesson is that a local freshwater relationship can be detectable without explaining most year-to-year adult returns, which integrate many life stages and management effects. Longer continuous creek-temperature records and better release, marine-survival, and basin-specific hydrology data would make the next test much stronger.

ACKNOWLEDGMENT.

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DATA AND CODE.

Project code, processed data products, Phase 7 result tables, and validation scripts are available at https://github.com/mtnh1gh/issaquah-salmon-study.

FIGURES.

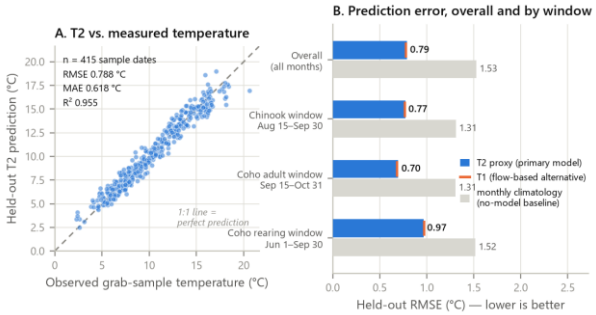


Figure 1. Leave-one-year-out validation of the primary T2 water-temperature reconstruction. T2 excludes streamflow so flow can later be tested independently against modeled temperature. The monthly-climatology bars provide a no-model baseline.

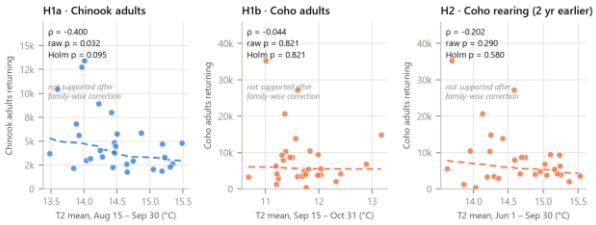


Figure 2. Prespecified salmon-temperature associations. Chinook adult migration (left) showed a moderate negative association, while adult Coho migration (center) and Coho juvenile/rearing exposure two years earlier (right) were weaker. Dashed lines are descriptive fits; confirmatory inference used Spearman correlation and permutation tests.

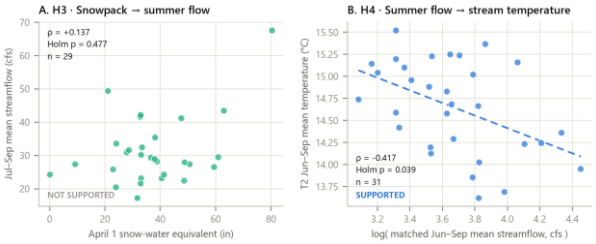


Figure 3. Physical-mechanism tests. April 1 Stampede Pass snow-water equivalent was not clearly associated with July–September Issaquah Creek flow (left). Higher matched June–September flow was associated with cooler modeled creek temperature (right).

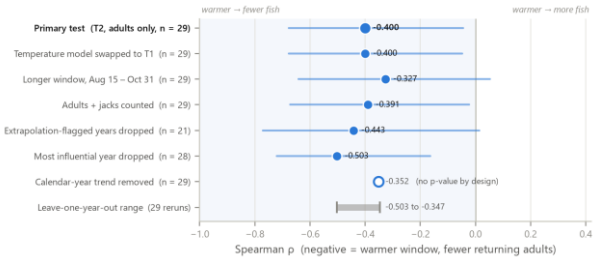


Figure 4. Prespecified robustness checks for the Chinook A1 association. The relationship remained negative across alternative temperature models, windows, outcomes, exclusion rules, and every leave-one-year-out rerun.

REFERENCES.

[1] D. A. McCullough, A Review and Synthesis of Effects of Alterations to the Water Temperature Regime on Freshwater Life Stages of Salmonids, with Special Reference to Chinook Salmon. U.S. EPA, EPA 910-R-99-010, 1999.

[2] U.S. Environmental Protection Agency, EPA Region 10 Guidance for Pacific Northwest State and Tribal Temperature Water Quality Standards, EPA 910-B-03-002, 2003.

[3] A. Richter and S. A. Kolmes, "Maximum temperature limits for Chinook, Coho, and chum salmon, and steelhead trout in the Pacific Northwest," Reviews in Fisheries Science, vol. 13, no. 1, pp. 23–49, 2005, doi:10.1080/10641260590885861.

[4] J. C. Newell, K. L. Fresh, and T. P. Quinn, "Arrival patterns and movements of adult sockeye salmon in Lake Washington: Implications for management of an urban fishery," North American Journal of Fisheries Management, vol. 27, no. 3, pp. 908–917, 2007, doi:10.1577/M06-058.1.

[5] H. K. Barnett, T. P. Quinn, M. Bhuthimethee, and J. Winton, "Increased prespawning mortality threatens an integrated natural- and hatchery-origin sockeye salmon population in the Lake Washington Basin," Fisheries Research, vol. 227, 105527, 2020, doi:10.1016/j.fishres.2020.105527.

[6] T. P. Quinn, J. Ray, and M. D. Tillotson, "Despite warming temperatures, hatchery origin Chinook and coho salmon spawn earlier in the fall: A multi-decade analysis," Hydrobiologia, 2026, doi:10.1007/s10750-026-06308-7.

[7] Washington Department of Fish and Wildlife, "Hatchery Adult Salmon Returns, Issaquah Hatchery / Issaquah Creek stock," Washington State Open Data, accessed July 19, 2026.

[8] King County, "Water Quality Data—Issaquah Creek at SE 56th St (station 0631)," accessed Aug. 23, 2026.

[9] NOAA National Centers for Environmental Information, "Global Historical Climatology Network-Daily, station USW00024233," accessed Aug. 23, 2026.

[10] U.S. Geological Survey, "National Water Information System, site 12121600, Issaquah Creek near mouth near Issaquah, WA," accessed 2026.

[11] USDA Natural Resources Conservation Service, "SNOTEL station 788, Stampede Pass, Washington," accessed July 19, 2026.

[12] D. Caissie, "The thermal regime of rivers: A review," Freshwater Biology, vol. 51, no. 8, pp. 1389–1406, 2006, doi:10.1111/j.1365-2427.2006.01597.x.

[13] L. E. Stratton Garvin, S. A. Rounds, and N. L. Buccola, Estimating Stream Temperature in the Willamette River Basin, Northwestern Oregon: A Regression-Based Approach, U.S. Geological Survey Scientific Investigations Report 2021–5022, 2022, doi:10.3133/sir20215022.

[14] G. S. Mauger et al., State of Knowledge: Climate Change in Puget Sound. Climate Impacts Group, University of Washington, 2015, doi:10.7915/CIG93777D.